

# Overview in General

This file contains a general overview of the data in the graph including node labels and relationships types.

## References

- [jqassistant](#)
- [Neo4j Python Driver](#)

## Node Labels

### Table 1a - Highest node count by label combination

Lists the 30 label combinations with the highest number of nodes. The labels with the lowest node count are listed in table 1b. The total list would sum up to the total number of labels (100%).

The whole table can be found in the CSV report `Node_label_combination_count`.

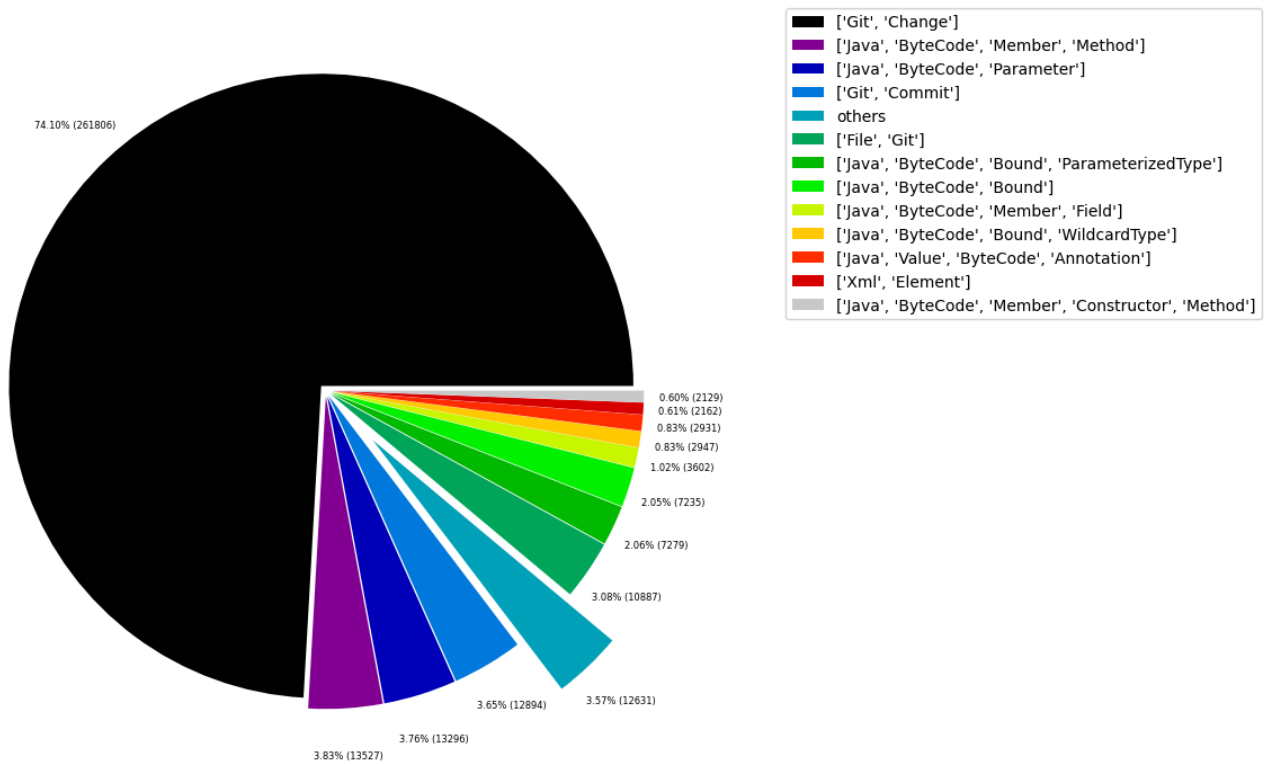
|    | nodeLabels                                        | nodesWithThatLabels | nodesWithThatLabelsPercent |
|----|---------------------------------------------------|---------------------|----------------------------|
| 0  | [Git, Change]                                     | 261806              | 74.097576                  |
| 1  | [Java, ByteCode, Member, Method]                  | 13527               | 3.828476                   |
| 2  | [Java, ByteCode, Parameter]                       | 13296               | 3.763097                   |
| 3  | [Git, Commit]                                     | 12894               | 3.649321                   |
| 4  | [File, Git]                                       | 10887               | 3.081290                   |
| 5  | [Java, ByteCode, Bound, ParameterizedType]        | 7279                | 2.060137                   |
| 6  | [Java, ByteCode, Bound]                           | 7235                | 2.047684                   |
| 7  | [Java, ByteCode, Member, Field]                   | 3602                | 1.019455                   |
| 8  | [Java, ByteCode, Bound, WildcardType]             | 2947                | 0.834074                   |
| 9  | [Java, Value, ByteCode, Annotation]               | 2931                | 0.829546                   |
| 10 | [Xml, Element]                                    | 2162                | 0.611899                   |
| 11 | [Java, ByteCode, Member, Constructor, Method]     | 2129                | 0.602560                   |
| 12 | [Xml, Text]                                       | 1450                | 0.410386                   |
| 13 | [Java, ByteCode, TypeVariable, Bound]             | 1111                | 0.314440                   |
| 14 | [Java, ByteCode, Member, Method, Lambda]          | 972                 | 0.275100                   |
| 15 | [Type, File, Java, ByteCode, ResolvedDuplicate... | 888                 | 0.251326                   |
| 16 | [Type, File, Java, Class, ByteCode]               | 846                 | 0.239439                   |
| 17 | [Json, Key]                                       | 702                 | 0.198683                   |
| 18 | [Value, Json, Scalar]                             | 685                 | 0.193872                   |
| 19 | [Java, Value, ByteCode, Primitive]                | 672                 | 0.190193                   |
| 20 | [Type, File, Java, ByteCode, JavaType]            | 649                 | 0.183683                   |
| 21 | [Java, ByteCode, Member, Method, GenericDeclar... | 578                 | 0.163588                   |
| 22 | [Type, File, Java, ByteCode, ExternalType]        | 411                 | 0.116323                   |
| 23 | [Author, Git, Person]                             | 298                 | 0.084341                   |
| 24 | [Value, Array]                                    | 287                 | 0.081228                   |
| 25 | [Committer, Git, Person]                          | 252                 | 0.071322                   |
| 26 | [Type, File, Java, Class, ByteCode, GenericDec... | 237                 | 0.067077                   |
| 27 | [Value, Property]                                 | 211                 | 0.059718                   |
| 28 | [Type, File, Java, ByteCode, Interface]           | 190                 | 0.053775                   |
| 29 | [Java, Value, Class, ByteCode]                    | 189                 | 0.053492                   |

## Chart 1a - Highest node count by label combination

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in Chart 1b.

<Figure size 640x480 with 0 Axes>

Nodes per label combination (more than 0.5% overall)



## Table 1b - Lowest node count by label combination

Lists the 30 label combinations with the lowest number of nodes until they reach 0.5% of the total node count, which are shown above.

|    | nodeLabels                                         | nodesWithThatLabels | nodesWithThatLabelsPercent |
|----|----------------------------------------------------|---------------------|----------------------------|
| 0  | [Analyze, Task, jQAssistant]                       | 1                   | 0.000283                   |
| 1  | [Package, File, Json, NPM]                         | 1                   | 0.000283                   |
| 2  | [Repository, File, Git]                            | 1                   | 0.000283                   |
| 3  | [File, TS, Scan]                                   | 1                   | 0.000283                   |
| 4  | [File, Json]                                       | 2                   | 0.000566                   |
| 5  | [File]                                             | 3                   | 0.000849                   |
| 6  | [Java, ByteCode, Member, Constructor, Method, ...] | 4                   | 0.001132                   |
| 7  | [Maven, Exclusion]                                 | 5                   | 0.001415                   |
| 8  | [Value, Array, Json]                               | 6                   | 0.001698                   |
| 9  | [Dependency, NPM]                                  | 7                   | 0.001981                   |
| 10 | [Type, File, Java, ByteCode, Void]                 | 9                   | 0.002547                   |
| 11 | [File, Maven, Xml, Pom, Document]                  | 9                   | 0.002547                   |
| 12 | [Java, ManifestSection]                            | 9                   | 0.002547                   |
| 13 | [File, Java, Manifest]                             | 9                   | 0.002547                   |
| 14 | [Artifact, File, Jar, Archive, Zip, Java]          | 9                   | 0.002547                   |
| 15 | [File, Java, ServiceLoader]                        | 10                  | 0.002830                   |
| 16 | [File, Java, Properties]                           | 12                  | 0.003396                   |
| 17 | [Maven, PluginExecution]                           | 16                  | 0.004528                   |
| 18 | [Maven, ExecutionGoal]                             | 16                  | 0.004528                   |
| 19 | [Xml, Attribute]                                   | 18                  | 0.005094                   |
| 20 | [jQAssistant, Rule, Concept]                       | 19                  | 0.005377                   |
| 21 | [Maven, Configuration]                             | 21                  | 0.005944                   |
| 22 | [Maven, Plugin]                                    | 21                  | 0.005944                   |
| 23 | [Type, File, Java, ByteCode, Throwable, Extern...  | 21                  | 0.005944                   |
| 24 | [Type, File, Java, ByteCode, Throwable, Resolv...  | 22                  | 0.006227                   |
| 25 | [Type, File, Java, ByteCode, Enum]                 | 28                  | 0.007925                   |
| 26 | [Type, File, Java, ByteCode, PrimitiveType]        | 30                  | 0.008491                   |
| 27 | [Xml, Namespace]                                   | 36                  | 0.010189                   |
| 28 | [Git, Branch]                                      | 43                  | 0.012170                   |
| 29 | [Type, File, Java, ByteCode, Annotation]           | 44                  | 0.012453                   |

## Chart 1b - Lowest node count by label combination

Shows the lowest (less than 0.5% overall) node count label combinations. Therefore, this plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

<Figure size 640x480 with 0 Axes>

Nodes per label combination (less than 0.5% overall)

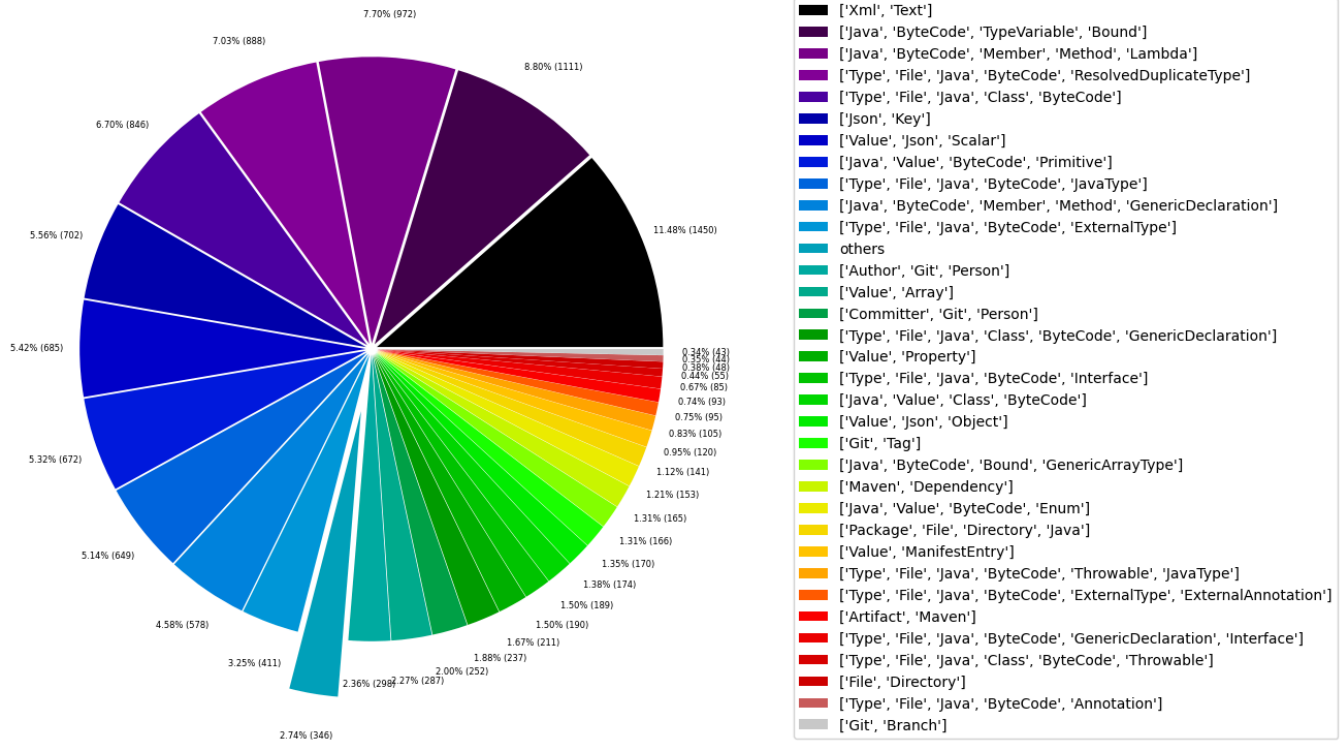


Table 1c - Highest node count by single label

Lists the 40 labels with the highest number of nodes. Doesn't sum up to the total number of nodes or 100% because one node can have multiple labels. Helps to identify commonly used labels.

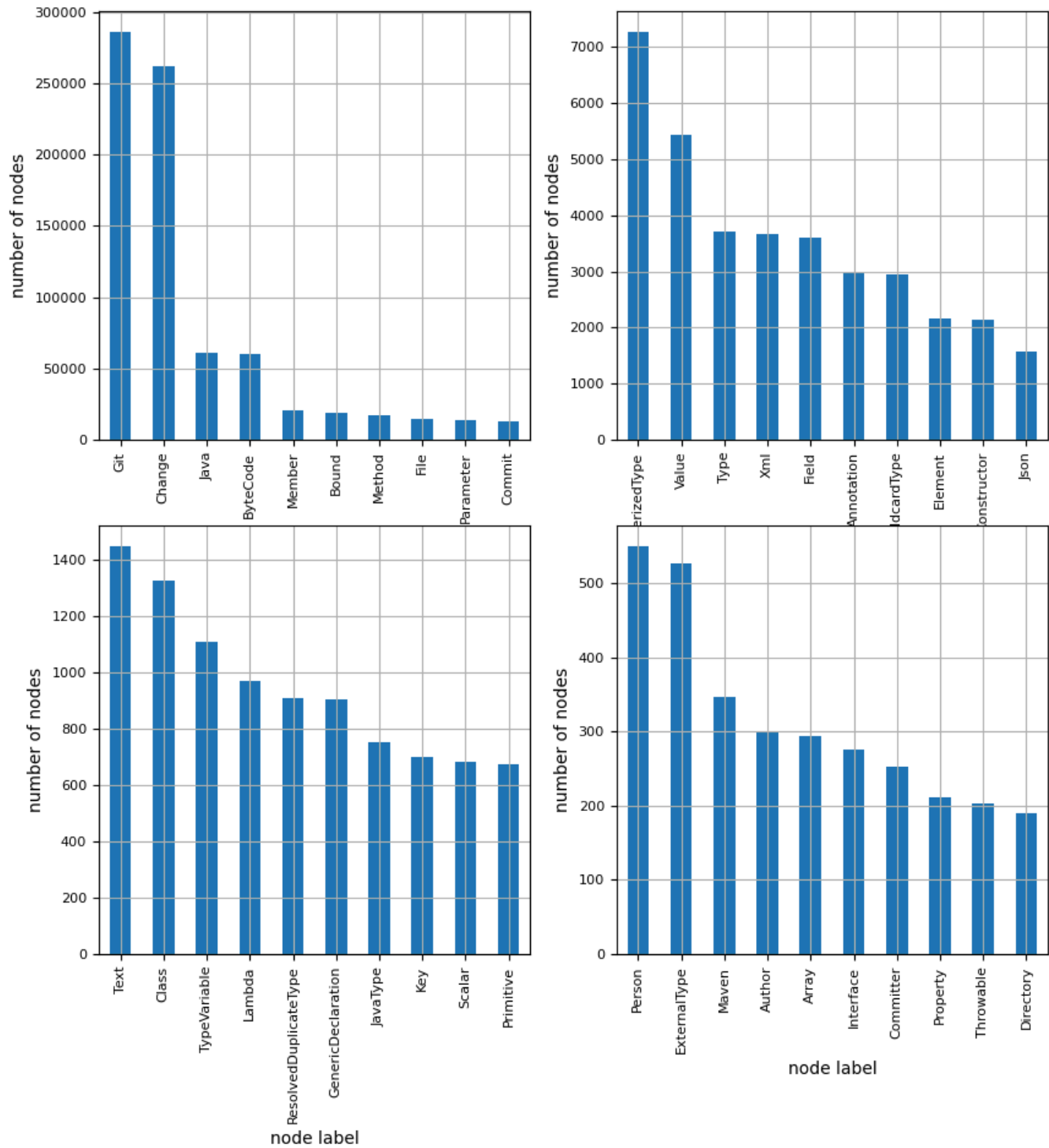
|    | nodeLabel             | nodesWithThatLabel | nodesWithThatLabelPercent |
|----|-----------------------|--------------------|---------------------------|
| 0  | Git                   | 286351             | 81.044418                 |
| 1  | Change                | 261806             | 74.097576                 |
| 2  | Java                  | 60696              | 17.178470                 |
| 3  | ByteCode              | 60506              | 17.124695                 |
| 4  | Member                | 20812              | 5.890311                  |
| 5  | Bound                 | 18738              | 5.303318                  |
| 6  | Method                | 17210              | 4.870856                  |
| 7  | File                  | 14848              | 4.202351                  |
| 8  | Parameter             | 13296              | 3.763097                  |
| 9  | Commit                | 12894              | 3.649321                  |
| 10 | ParameterizedType     | 7279               | 2.060137                  |
| 11 | Value                 | 5428               | 1.536258                  |
| 12 | Type                  | 3715               | 1.051437                  |
| 13 | Xml                   | 3675               | 1.040116                  |
| 14 | Field                 | 3602               | 1.019455                  |
| 15 | Annotation            | 2975               | 0.841999                  |
| 16 | WildcardType          | 2947               | 0.834074                  |
| 17 | Element               | 2162               | 0.611899                  |
| 18 | Constructor           | 2133               | 0.603692                  |
| 19 | Json                  | 1570               | 0.444349                  |
| 20 | Text                  | 1450               | 0.410386                  |
| 21 | Class                 | 1327               | 0.375574                  |
| 22 | TypeVariable          | 1111               | 0.314440                  |
| 23 | Lambda                | 972                | 0.275100                  |
| 24 | ResolvedDuplicateType | 910                | 0.257553                  |
| 25 | GenericDeclaration    | 904                | 0.255854                  |
| 26 | JavaType              | 754                | 0.213401                  |
| 27 | Key                   | 702                | 0.198683                  |
| 28 | Scalar                | 685                | 0.193872                  |
| 29 | Primitive             | 672                | 0.190193                  |
| 30 | Person                | 550                | 0.155664                  |
| 31 | ExternalType          | 527                | 0.149154                  |
| 32 | Maven                 | 346                | 0.097927                  |
| 33 | Author                | 298                | 0.084341                  |
| 34 | Array                 | 293                | 0.082926                  |
| 35 | Interface             | 275                | 0.077832                  |
| 36 | Committer             | 252                | 0.071322                  |
| 37 | Property              | 211                | 0.059718                  |
| 38 | Throwable             | 203                | 0.057454                  |
| 39 | Directory             | 189                | 0.053492                  |

## Chart 1c - Highest node count by label

Shows the 40 labels with the highest number of nodes.

<Figure size 640x480 with 0 Axes>

## Node count by label



## Relationship Types

Table 2a - Highest relationship count by type

Lists the 30 relationship types with the highest number of occurrences. The whole table can be found in the CSV report `Relationship_type_count`.

Total number of relationships: 1096486

|    | relationshipType         | nodesWithThatRelationshipType | nodesWithThatRelationshipTypePercent |
|----|--------------------------|-------------------------------|--------------------------------------|
| 0  | CONTAINS_CHANGE          | 261806                        | 23.876821                            |
| 1  | MODIFIES                 | 261806                        | 23.876821                            |
| 2  | UPDATES                  | 172624                        | 15.743384                            |
| 3  | CREATES                  | 61715                         | 5.628435                             |
| 4  | DELETES                  | 38029                         | 3.468261                             |
| 5  | INVOKES                  | 36628                         | 3.340490                             |
| 6  | COMMITTED                | 25788                         | 2.351877                             |
| 7  | DEPENDS_ON               | 22488                         | 2.050915                             |
| 8  | OF_TYPE                  | 21922                         | 1.999296                             |
| 9  | DECLARES                 | 21273                         | 1.940107                             |
| 10 | OF_RAW_TYPE              | 17348                         | 1.582145                             |
| 11 | HAS_PARENT               | 15607                         | 1.423365                             |
| 12 | HAS                      | 14432                         | 1.316205                             |
| 13 | HAS_COMMIT               | 12894                         | 1.175938                             |
| 14 | RETURNS                  | 12853                         | 1.172199                             |
| 15 | HAS_FILE                 | 10887                         | 0.992899                             |
| 16 | RENAMES                  | 10562                         | 0.963259                             |
| 17 | READS                    | 9378                          | 0.855278                             |
| 18 | HAS_ACTUAL_TYPE_ARGUMENT | 8407                          | 0.766722                             |
| 19 | HAS_NEW_NAME             | 6240                          | 0.569091                             |
| 20 | OF_GENERIC_TYPE          | 5991                          | 0.546382                             |
| 21 | RESOLVES_TO              | 5277                          | 0.481265                             |
| 22 | SIMILAR                  | 4078                          | 0.371915                             |
| 23 | WRITES                   | 3944                          | 0.359695                             |
| 24 | CONTAINS                 | 3936                          | 0.358965                             |
| 25 | RETURNS_GENERIC          | 3593                          | 0.327683                             |
| 26 | ANNOTATED_BY             | 2919                          | 0.266214                             |
| 27 | REQUIRES                 | 2230                          | 0.203377                             |
| 28 | HAS_FIRST_CHILD          | 2162                          | 0.197175                             |
| 29 | HAS_LAST_CHILD           | 2162                          | 0.197175                             |

## Chart 2a - Highest relationship count by type

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in the second chart.

<Figure size 640x480 with 0 Axes>



Relationship types (more than 0.5% overall)

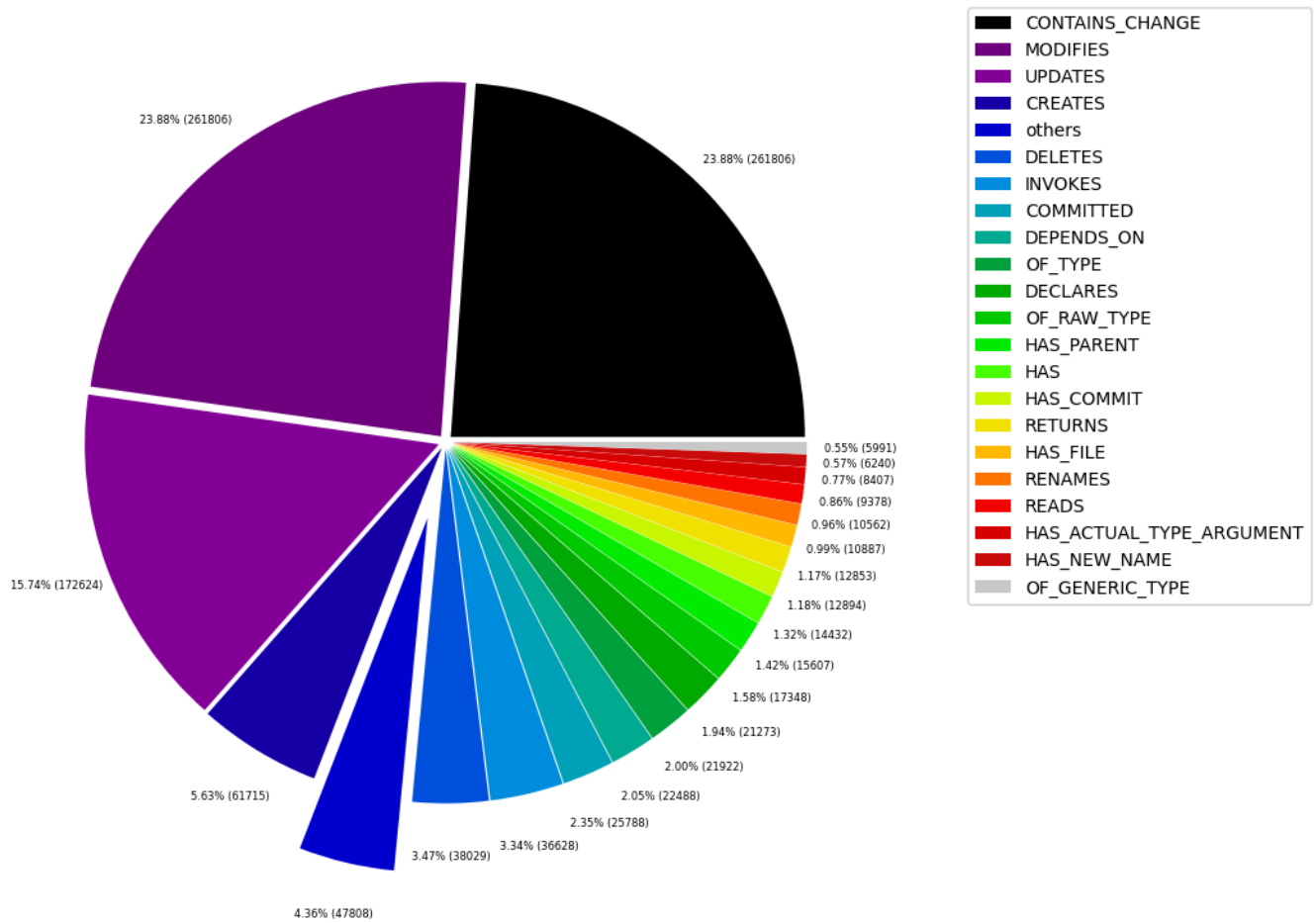


Table 2b - Lowest relationship count by type

Lists the 30 relationships type with the lowest number of occurrences up to 0.5% of the total node count. This is essentially breaking down the "others" slice from the chart above.

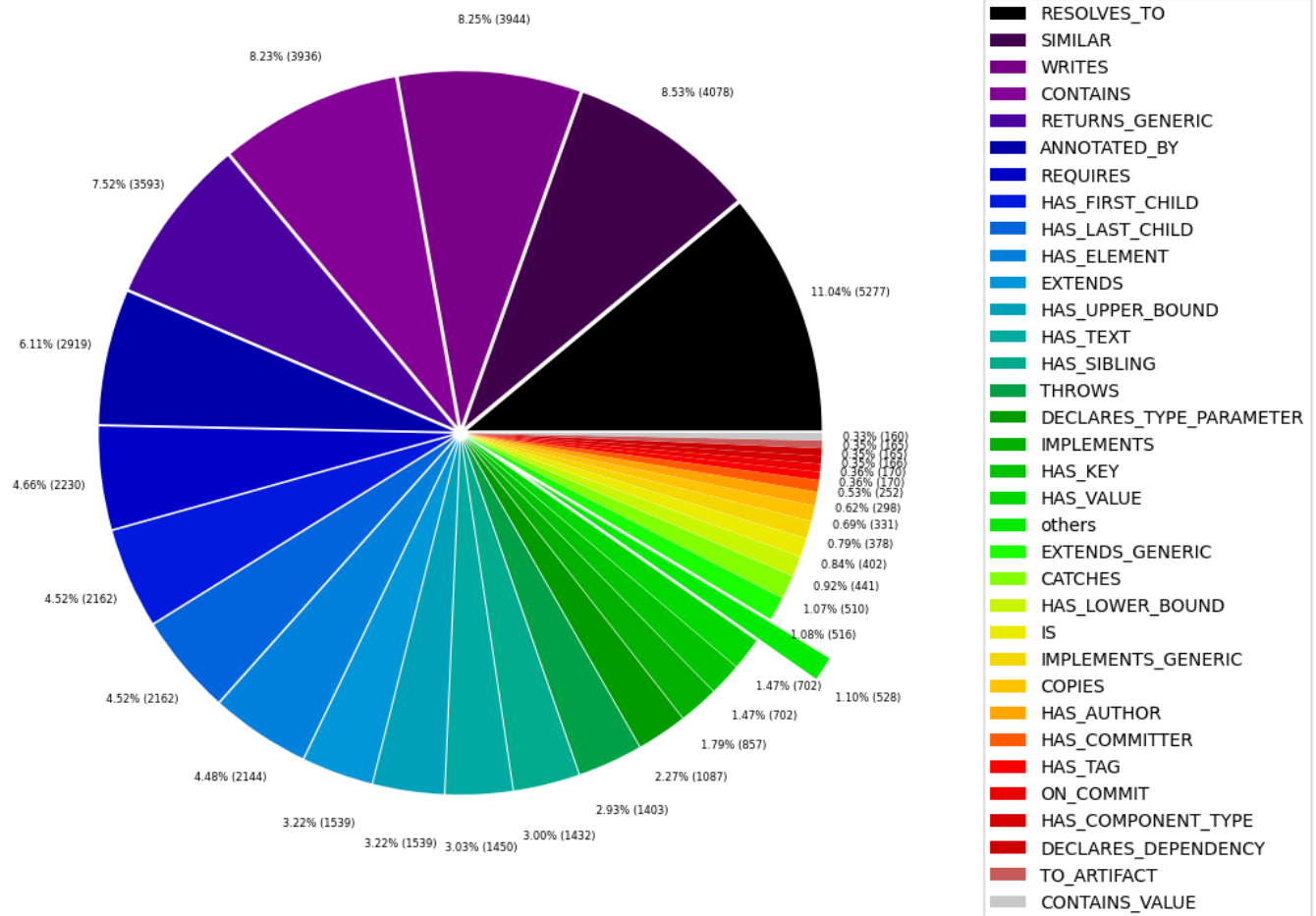
|    | relationshipType        | nodesWithThatRelationshipType | nodesWithThatRelationshipTypePercent |
|----|-------------------------|-------------------------------|--------------------------------------|
| 0  | HAS_PROPERTY            | 1                             | 0.000091                             |
| 1  | THROWS_GENERIC          | 5                             | 0.000456                             |
| 2  | EXCLUDES                | 5                             | 0.000456                             |
| 3  | DECLARES_DEV_DEPENDENCY | 7                             | 0.000638                             |
| 4  | DESCRIBES               | 9                             | 0.000821                             |
| 5  | HAS_ROOT_ELEMENT        | 11                            | 0.001003                             |
| 6  | HAS_GOAL                | 16                            | 0.001459                             |
| 7  | HAS_EXECUTION           | 16                            | 0.001459                             |
| 8  | OF_NAMESPACE            | 18                            | 0.001642                             |
| 9  | HAS_ATTRIBUTE           | 18                            | 0.001642                             |
| 10 | INCLUDES_CONCEPT        | 19                            | 0.001733                             |
| 11 | IS_ARTIFACT             | 21                            | 0.001915                             |
| 12 | HAS_CONFIGURATION       | 21                            | 0.001915                             |
| 13 | USES_PLUGIN             | 21                            | 0.001915                             |
| 14 | REQUIRES_TYPE_PARAMETER | 24                            | 0.002189                             |
| 15 | REQUIRES_CONCEPT        | 28                            | 0.002554                             |
| 16 | DECLARES_NAMESPACE      | 36                            | 0.003283                             |
| 17 | HAS_DEFAULT             | 39                            | 0.003557                             |
| 18 | HAS_BRANCH              | 43                            | 0.003922                             |
| 19 | HAS_HEAD                | 44                            | 0.004013                             |
| 20 | COPY_OF                 | 126                           | 0.011491                             |
| 21 | CONTAINS_VALUE          | 160                           | 0.014592                             |
| 22 | DECLARES_DEPENDENCY     | 165                           | 0.015048                             |
| 23 | TO_ARTIFACT             | 165                           | 0.015048                             |
| 24 | HAS_COMPONENT_TYPE      | 166                           | 0.015139                             |
| 25 | ON_COMMIT               | 170                           | 0.015504                             |
| 26 | HAS_TAG                 | 170                           | 0.015504                             |
| 27 | HAS_COMMITTER           | 252                           | 0.022983                             |
| 28 | HAS_AUTHOR              | 298                           | 0.027178                             |
| 29 | COPIES                  | 331                           | 0.030187                             |

## Chart 2b - Lowest relationship count by type

Shows the lowest (less than 0.5% overall) relationship types. This plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

<Figure size 640x480 with 0 Axes>

Relationship types (less than 0.5% overall)



## Node labels with their relationships

Table 3a - Highest relationship count by node labels and relationship type

Lists the 30 node labels and their relationship types with the highest number of occurrences.

|    | sourceLabels                                  | relationType             | targetLabels                                      | numberOfRelationships | numberOfNodesWithSameLabelsAsSource | numbe |
|----|-----------------------------------------------|--------------------------|---------------------------------------------------|-----------------------|-------------------------------------|-------|
| 0  | [Git, Change]                                 | MODIFIES                 | [File, Git]                                       | 261806                | 261806                              |       |
| 1  | [Git, Commit]                                 | CONTAINS_CHANGE          | [Git, Change]                                     | 261806                | 12894                               |       |
| 2  | [Git, Change]                                 | UPDATES                  | [File, Git]                                       | 172624                | 261806                              |       |
| 3  | [Git, Change]                                 | CREATES                  | [File, Git]                                       | 61715                 | 261806                              |       |
| 4  | [Git, Change]                                 | DELETES                  | [File, Git]                                       | 38029                 | 261806                              |       |
| 5  | [Java, ByteCode, Member, Method]              | INVOKES                  | [Java, ByteCode, Member, Method]                  | 22255                 | 13428                               |       |
| 6  | [Git, Commit]                                 | HAS_PARENT               | [Git, Commit]                                     | 15598                 | 12894                               |       |
| 7  | [Repository, File, Git]                       | HAS_COMMIT               | [Git, Commit]                                     | 12894                 | 1                                   |       |
| 8  | [Author, Git, Person]                         | COMMITTED                | [Git, Commit]                                     | 12894                 | 298                                 |       |
| 9  | [Committer, Git, Person]                      | COMMITTED                | [Git, Commit]                                     | 12894                 | 252                                 |       |
| 10 | [Repository, File, Git]                       | HAS_FILE                 | [File, Git]                                       | 10887                 | 1                                   |       |
| 11 | [Git, Change]                                 | RENAMES                  | [File, Git]                                       | 10562                 | 261806                              |       |
| 12 | [Java, ByteCode, Member, Method]              | HAS                      | [Java, ByteCode, Parameter]                       | 8495                  | 13428                               |       |
| 13 | [Java, ByteCode, Member, Method]              | READS                    | [Java, ByteCode, Member, Field]                   | 8420                  | 13428                               |       |
| 14 | [File, Git]                                   | HAS_NEW_NAME             | [File, Git]                                       | 6240                  | 10887                               |       |
| 15 | [Java, ByteCode, Parameter]                   | OF_TYPE                  | [Type, File, Java, ByteCode, JavaType]            | 6160                  | 13296                               |       |
| 16 | [Java, ByteCode, Bound]                       | OF_RAW_TYPE              | [Type, File, Java, ByteCode, JavaType]            | 3465                  | 7235                                |       |
| 17 | [Java, ByteCode, Bound, ParameterizedType]    | OF_RAW_TYPE              | [Type, File, Java, ByteCode, JavaType]            | 3110                  | 7279                                |       |
| 18 | [Java, ByteCode, Bound, ParameterizedType]    | HAS_ACTUAL_TYPE_ARGUMENT | [Java, ByteCode, Bound, WildcardType]             | 2947                  | 7279                                |       |
| 19 | [Java, ByteCode, Parameter]                   | OF_GENERIC_TYPE          | [Java, ByteCode, Bound, ParameterizedType]        | 2695                  | 13296                               |       |
| 20 | [Java, ByteCode, Member, Constructor, Method] | WRITES                   | [Java, ByteCode, Member, Field]                   | 2519                  | 2129                                |       |
| 21 | [Java, ByteCode, Bound, ParameterizedType]    | HAS_ACTUAL_TYPE_ARGUMENT | [Java, ByteCode, TypeVariable, Bound]             | 2474                  | 7279                                |       |
| 22 | [Java, Value, ByteCode, Annotation]           | OF_TYPE                  | [Type, File, Java, ByteCode, ExternalType, Ext... | 2445                  | 2931                                |       |
| 23 | [Java, ByteCode, Member, Constructor, Method] | INVOKES                  | [Java, ByteCode, Member, Constructor, Method]     | 2195                  | 2129                                |       |
| 24 | [Xml, Element]                                | HAS_ELEMENT              | [Xml, Element]                                    | 2144                  | 2162                                |       |
| 25 | [Java, ByteCode, Member, Method]              | INVOKES                  | [Java, ByteCode, Member, Constructor, Method]     | 2113                  | 13428                               |       |
| 26 | [Java, ByteCode, Member, Method]              | RETURNS                  | [Type, File, Java, ByteCode, JavaType]            | 2107                  | 13428                               |       |
| 27 | [Java, ByteCode, Member, Constructor, Method] | HAS                      | [Java, ByteCode, Parameter]                       | 2075                  | 2129                                |       |
| 28 | [Java, ByteCode, Parameter]                   | OF_GENERIC_TYPE          | [Java, ByteCode, Bound]                           | 2056                  | 13296                               |       |
| 29 | [Java, ByteCode, Member, Method, Lambda]      | INVOKES                  | [Java, ByteCode, Member, Method]                  | 1954                  | 972                                 |       |

## Graph Density

```
total_number_of_nodes (vertices): 353326
total_number_of_relationships (edges): 1096486
-> total directed graph density: 8.783207168521157e-06
-> total directed graph density in percent: 0.0008783207168521157
```